



Analog CMOS Circuit Design SS26

Practical Exercise V – Circuit Specifications

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Feedback

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Practical Exercise V – Answers

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Given values

$$\begin{aligned}C_L &= 3 \text{ pF}, \\ \omega_u &= 2\pi \cdot 20 \text{ MHz} = 125.66 \times 10^6 \text{ rad/s}, \\ SR &= 15 \text{ V}/\mu\text{s}, \\ V_{DD} &= 3.3 \text{ V}, \quad V_{SS} = 0 \text{ V}, \\ I_{REF} &= 10 \mu\text{A}, \\ \frac{C_1}{C_M} &\approx 0.3, \quad C_2 \approx C_L.\end{aligned}$$

T1: First stage differential amplifier

a) Miller capacitor

$$\begin{aligned}z_1 &= \frac{g_{m6}}{C_M} > 10\omega_u, \\ |p_2| &= \frac{g_{m6}}{C_2 \left(1 + \frac{C_1}{C_M}\right)} > 2.2\omega_u.\end{aligned}$$

$$\begin{aligned}\frac{z_1}{|p_2|} &= \frac{10}{2.2} = \frac{C_2 \left(1 + \frac{C_1}{C_M}\right)}{C_M}, \\ C_M &= \frac{2.2}{10} (1 + 0.3) C_2, \\ C_M &= 0.286 C_2.\end{aligned}$$

With $C_2 \approx C_L = 3 \text{ pF}$:

$$\boxed{C_M = 0.858 \text{ pF}}.$$

b) Tail current from slew rate

$$\begin{aligned}I_{5,min} &= C_M \cdot SR, \\ I_{5,min} &= 0.858 \text{ pF} \cdot 15 \text{ V}/\mu\text{s}, \\ I_{5,min} &= 12.87 \mu\text{A}.\end{aligned}$$

Chosen value:

$$\boxed{I_5 = 15 \mu\text{A}}.$$

Therefore:

$$\boxed{I_1 = I_2 = \frac{I_5}{2} = 7.5 \mu\text{A}}.$$

c) Input pair T_1, T_2

$$g_{m1,2} = \omega_u C_M,$$

$$g_{m1,2} = 2\pi \cdot 20 \text{ MHz} \cdot 0.858 \text{ pF},$$

$$\boxed{g_{m1,2}} = \boxed{107.82 \mu\text{S}}.$$

$$\left(\frac{g_m}{I_D}\right)_{1,2} = \frac{107.82 \mu\text{S}}{7.5 \mu\text{A}},$$

$$\boxed{\left(\frac{g_m}{I_D}\right)_{1,2}} = \boxed{14.38/\text{V}}.$$

Using the g_m/I_D lookup table for nMOS, $L_{1,2} = 1 \mu\text{m}$:

$$\left(\frac{I_D}{W}\right)_{1,2} = 1.0193 \text{ A/m},$$

$$W_{1,2} = \frac{7.5 \mu\text{A}}{1.0193 \text{ A/m}},$$

$$\boxed{W_1 = W_2} = \boxed{7.36 \mu\text{m}},$$

$$\boxed{L_1 = L_2} = \boxed{1 \mu\text{m}}.$$

$$\boxed{V_{GS1,2} = 0.609 \text{ V}}.$$

d) Tail current source T_5

$$V_{CM,min} = V_{SS} + 1 \text{ V} = 1 \text{ V},$$

$$V_{GS1,eff} = V_{GS1} + 0.10 \text{ V} = 0.709 \text{ V},$$

$$V_{DS5} = V_{CM,min} - V_{GS1,eff},$$

$$V_{DS5} = 1 \text{ V} - 0.709 \text{ V} = 0.291 \text{ V}.$$

$$V_{DS5} \geq \frac{2}{(g_m/I_D)_5},$$

$$\left(\frac{g_m}{I_D}\right)_{5,min} = \frac{2}{0.291 \text{ V}},$$

$$\left(\frac{g_m}{I_D}\right)_{5,min} = 6.88/\text{V}.$$

Chosen value:

$$\boxed{\left(\frac{g_m}{I_D}\right)_5 = 7/\text{V}}.$$

Using nMOS lookup, $L_5 = 2\text{ }\mu\text{m}$:

$$\begin{aligned}\left(\frac{I_D}{W}\right)_5 &= 2.9165\text{ A/m}, \\ W_5 &= \frac{15\text{ }\mu\text{A}}{2.9165\text{ A/m}}, \\ \boxed{W_5} &= \boxed{5.14\text{ }\mu\text{m}}, \\ \boxed{L_5} &= \boxed{2\text{ }\mu\text{m}}, \\ \boxed{(W/L)_5} &= \boxed{2.57}.\end{aligned}$$

e) PMOS load transistors T_3, T_4

$$\begin{aligned}V_{CM,max} &= V_{DD} - 1\text{ V} = 2.3\text{ V}, \\ V_{GS1,eff} &= V_{GS1} + 0.320\text{ V} = 0.929\text{ V}, \\ V_{S1} &= V_{CM,max} - V_{GS1,eff}, \\ V_{S1} &= 2.3\text{ V} - 0.929\text{ V} = 1.371\text{ V}.\end{aligned}$$

$$\begin{aligned}V_{DSsat1} &= \frac{2}{(g_m/I_D)_1} = \frac{2}{14.38} = 0.139\text{ V}, \\ V_{D1,min} &= V_{S1} + V_{DSsat1}, \\ V_{D1,min} &= 1.371\text{ V} + 0.139\text{ V} = 1.510\text{ V}.\end{aligned}$$

Since $V_{D1} = V_{D3} = V_{G3}$:

$$\begin{aligned}V_{G3,min} &= 1.510\text{ V}, \\ V_{SG3,max} &= V_{DD} - V_{G3,min} = 1.790\text{ V}.\end{aligned}$$

Chosen value:

$$\boxed{\left(\frac{g_m}{I_D}\right)_{3,4} = 1.5/\text{V}}.$$

Using pMOS lookup, $I_3 = I_4 = 7.5\text{ }\mu\text{A}$, $L_{3,4} = 2\text{ }\mu\text{m}$:

$$\begin{aligned}\left(\frac{I_D}{W}\right)_{3,4} &= 10.2129\text{ A/m}, \\ W_{3,4} &= \frac{7.5\text{ }\mu\text{A}}{10.2129\text{ A/m}}, \\ \boxed{W_3 = W_4} &= \boxed{0.734\text{ }\mu\text{m}}, \\ \boxed{L_3 = L_4} &= \boxed{2\text{ }\mu\text{m}}, \\ \boxed{(W/L)_{3,4}} &= \boxed{0.367}.\end{aligned}$$

T2: Second stage

a) Output-stage current I_6

$$\begin{aligned}|p_2| &= \frac{g_{m6}}{C_2 \left(1 + \frac{C_1}{C_M}\right)} > 2.2\omega_u, \\ g_{m6} &= 2.2\omega_u C_2 \left(1 + \frac{C_1}{C_M}\right), \\ g_{m6} &= 2.2 \cdot 2\pi \cdot 20 \text{ MHz} \cdot 3 \text{ pF} \cdot 1.3, \\ \boxed{g_{m6}} &= \boxed{1.078 \text{ mS}}.\end{aligned}$$

$$\boxed{g_{m6} \approx 10g_{m1,2}}.$$

Chosen value:

$$\boxed{\left(\frac{g_m}{I_D}\right)_6 = \left(\frac{g_m}{I_D}\right)_4 = 1.5/\text{V}}.$$

$$\begin{aligned}I_6 &= \frac{g_{m6}}{(g_m/I_D)_6}, \\ I_6 &= \frac{1.078 \text{ mS}}{1.5/\text{V}}, \\ \boxed{I_6} &= \boxed{718.8 \text{ }\mu\text{A}}.\end{aligned}$$

b) PMOS second-stage transistor T_6

Using pMOS lookup, $L_6 = 1 \text{ }\mu\text{m}$:

$$\begin{aligned}\left(\frac{I_D}{W}\right)_6 &= 19.3157 \text{ A/m}, \\ W_6 &= \frac{718.8 \text{ }\mu\text{A}}{19.3157 \text{ A/m}}, \\ \boxed{W_6} &= \boxed{37.2 \text{ }\mu\text{m}}, \\ \boxed{L_6} &= \boxed{1 \text{ }\mu\text{m}}, \\ \boxed{(W/L)_6} &= \boxed{37.2}.\end{aligned}$$

c) NMOS current source T_7

$$\begin{aligned}I_7 &= I_6 = 718.8 \text{ }\mu\text{A}, \\ \frac{I_7}{I_5} &= \frac{718.8 \text{ }\mu\text{A}}{15 \text{ }\mu\text{A}} = 47.92.\end{aligned}$$

$$\begin{aligned}\left(\frac{W}{L}\right)_7 &= \frac{I_7}{I_5} \left(\frac{W}{L}\right)_5, \\ \left(\frac{W}{L}\right)_7 &= 47.92 \cdot 2.57, \\ \boxed{(W/L)_7} &= \boxed{123.2}.\end{aligned}$$

With $L_7 = 2\text{ }\mu\text{m}$:

$$\boxed{W_7} = \boxed{246.5\text{ }\mu\text{m}},$$

$$\boxed{L_7} = \boxed{2\text{ }\mu\text{m}}.$$

T3: Input-referred offset voltage

a) Relevant mismatch pairs

$$\boxed{T_1/T_2 \quad \text{and} \quad T_3/T_4.}$$

b) Standard deviation of input-referred offset

$$A_{VT0n} = 6.18\text{ mV }\mu\text{m},$$

$$A_{VT0p} = 4.632\text{ mV }\mu\text{m}.$$

For equal-sized matched pairs:

$$\sigma_{\Delta V_T} = \sqrt{2} \frac{A_{VT}}{\sqrt{WL}}.$$

Input pair contribution:

$$\sigma_{VOS,12} = \sqrt{2} \frac{6.18}{\sqrt{7.36 \cdot 1}},$$

$$\boxed{\sigma_{VOS,12}} = \boxed{3.22\text{ mV}}.$$

Load pair contribution:

$$\sigma_{\Delta V_{T34}} = \sqrt{2} \frac{4.632}{\sqrt{0.734 \cdot 2}},$$

$$\sigma_{\Delta V_{T34}} = 5.41\text{ mV}.$$

$$g_{m3,4} = I_3 \left(\frac{g_m}{I_D} \right)_{3,4},$$

$$g_{m3,4} = 7.5\text{ }\mu\text{A} \cdot 1.5/\text{V} = 11.25\text{ }\mu\text{S},$$

$$\frac{g_{m3,4}}{g_{m1,2}} = \frac{11.25\text{ }\mu\text{S}}{107.82\text{ }\mu\text{S}} = 0.104.$$

$$\sigma_{VOS,34} = \frac{g_{m3,4}}{g_{m1,2}} \sigma_{\Delta V_{T34}},$$

$$\sigma_{VOS,34} = 0.104 \cdot 5.41\text{ mV},$$

$$\boxed{\sigma_{VOS,34}} = \boxed{0.564\text{ mV}}.$$

Total:

$$\begin{aligned}\sigma_{VOS} &= \sqrt{\sigma_{VOS,12}^2 + \sigma_{VOS,34}^2}, \\ \sigma_{VOS} &= \sqrt{3.22^2 + 0.564^2} \text{ mV}, \\ \boxed{\sigma_{VOS}} &= \boxed{3.27 \text{ mV}}.\end{aligned}$$

$$3.27 \text{ mV} > 1.33 \text{ mV}.$$

Offset specification is not fulfilled.

c) Redesign for offset specification

Chosen redesigned lengths:

$$L_1 = L_2 = 3 \mu\text{m}, \quad L_3 = L_4 = 3 \mu\text{m}.$$

Same currents and same g_m/I_D ratios are kept.

Using g_m/I_D lookup:

$$\begin{aligned}\boxed{W_1 = W_2} &= \boxed{23.1 \mu\text{m}}, \\ \boxed{L_1 = L_2} &= \boxed{3 \mu\text{m}}, \\ \boxed{W_3 = W_4} &= \boxed{1.08 \mu\text{m}}, \\ \boxed{L_3 = L_4} &= \boxed{3 \mu\text{m}}.\end{aligned}$$

New offset values:

$$\begin{aligned}\sigma_{VOS,12,new} &= \sqrt{2} \frac{6.18}{\sqrt{23.1 \cdot 3}}, \\ \boxed{\sigma_{VOS,12,new}} &= \boxed{1.05 \text{ mV}}, \\ \sigma_{\Delta V_{T34},new} &= \sqrt{2} \frac{4.632}{\sqrt{1.08 \cdot 3}}, \\ \sigma_{\Delta V_{T34},new} &= 3.64 \text{ mV}, \\ \sigma_{VOS,34,new} &= 0.104 \cdot 3.64 \text{ mV}, \\ \boxed{\sigma_{VOS,34,new}} &= \boxed{0.380 \text{ mV}}.\end{aligned}$$

Total after redesign:

$$\begin{aligned}\sigma_{VOS,new} &= \sqrt{1.05^2 + 0.380^2} \text{ mV}, \\ \boxed{\sigma_{VOS,new}} &= \boxed{1.12 \text{ mV}}, \\ 3\sigma_{VOS,new} &= 3.35 \text{ mV}.\end{aligned}$$

$$1.12 \text{ mV} < 1.33 \text{ mV},$$

$$3.35 \text{ mV} < 4 \text{ mV}.$$

Offset specification is fulfilled after redesign.

Final transistor dimensions

Transistor	Type	Current	g_m/I_D	W	L
T_1, T_2	nMOS	7.5 μA	14.38/V	23.1 μm	3 μm
T_3, T_4	pMOS	7.5 μA	1.5/V	1.08 μm	3 μm
T_5	nMOS	15 μA	7/V	5.14 μm	2 μm
T_6	pMOS	718.8 μA	1.5/V	37.2 μm	1 μm
T_7	nMOS	718.8 μA	7/V	246.5 μm	2 μm

Final main values

$$\begin{aligned}
C_M &= 0.858 \text{ pF}, \\
I_5 &= 15 \text{ } \mu\text{A}, \\
g_{m1,2} &= 107.82 \text{ } \mu\text{S}, \\
g_{m6} &= 1.078 \text{ mS}, \\
I_6 = I_7 &= 718.8 \text{ } \mu\text{A}, \\
\sigma_{VOS,new} &= 1.12 \text{ mV}.
\end{aligned}$$